

Thermodynamics and Statistical Physics

Spring 2026

Homework 12

Issued: June 16, 2026

Due: **Sunday, June 21, 2026, 11:00 PM**

Student Information

Name: Xu Chen

Student ID: 523110910017

Email: xuchen2005@sjtu.edu.cn

Show all essential steps in your derivations.

Low- T expansion of a three-dimensional nonrelativistic Fermi gas

In this problem we systematically study the low- T expansion of a three-dimensional non-relativistic Fermi gas. Use the following equations, which have been derived in the lectures:

$$N = \int_0^\infty \frac{D(\epsilon) d\epsilon}{e^{\beta(\epsilon-\mu)} + 1}, \quad (1)$$

$$U = \int_0^\infty \frac{\epsilon D(\epsilon) d\epsilon}{e^{\beta(\epsilon-\mu)} + 1}, \quad (2)$$

$$D(\epsilon) d\epsilon = \frac{3N}{2\epsilon_F^{3/2}} \epsilon^{1/2} d\epsilon. \quad (3)$$

- Use Eq. (3) in Eq. (1), and apply the Sommerfeld expansion to obtain an expansion for $\epsilon_F^{3/2}$ in terms of T/μ . Compute the expansion through order $(T/\mu)^4$.
- Invert the series to obtain an expansion for μ in terms of T/ϵ_F , through order $(T/\epsilon_F)^4$. This is the low- T expansion of the chemical potential.
- Use the same method to obtain an expansion for U in terms of T/μ , through order $(T/\mu)^4$.
- Substitute the result of part (b) into the result of part (c), and obtain the low- T expansion of the internal energy through order $(T/\epsilon_F)^4$.
- Use the exact relation between P and U to obtain the low- T expansion of the pressure. Then compute the isothermal compressibility:

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{T,N} = \frac{1}{n} \left(\frac{\partial n}{\partial P} \right)_T. \quad (4)$$

Solution

- Substituting the density of states $D(\epsilon) = \frac{3N}{2\epsilon_F^{3/2}} \epsilon^{1/2}$ into the particle number equation gives:

$$N = \frac{3N}{2\epsilon_F^{3/2}} \int_0^\infty \frac{\epsilon^{1/2} d\epsilon}{e^{\beta(\epsilon-\mu)} + 1} \implies \epsilon_F^{3/2} = \frac{3}{2} \int_0^\infty \frac{\epsilon^{1/2} d\epsilon}{e^{\beta(\epsilon-\mu)} + 1}. \quad (5)$$

To evaluate this, we use the Sommerfeld expansion formula:

$$\int_0^\infty \frac{\phi(\epsilon) d\epsilon}{e^{\beta(\epsilon-\mu)} + 1} = \Phi(\mu) + \frac{\pi^2}{6} T^2 \Phi''(\mu) + \frac{7\pi^4}{360} T^4 \Phi^{(4)}(\mu) + O(T^6), \quad (6)$$

where $\Phi(\epsilon) = \int_0^\epsilon \phi(x) dx$ (setting $k_B = 1$ so $T \equiv k_B T$). For our integral, $\phi(\epsilon) = \epsilon^{1/2}$, so we have:

$$\Phi(\epsilon) = \frac{2}{3} \epsilon^{3/2}, \quad \Phi''(\epsilon) = \frac{1}{2} \epsilon^{-1/2}, \quad \Phi'''(\epsilon) = -\frac{1}{4} \epsilon^{-3/2}, \quad \Phi^{(4)}(\epsilon) = \frac{3}{8} \epsilon^{-5/2}. \quad (7)$$

Plugging these derivatives evaluated at μ into the expansion gives:

$$\begin{aligned}\epsilon_F^{3/2} &= \frac{3}{2} \left[\frac{2}{3} \mu^{3/2} + \frac{\pi^2}{6} T^2 \left(\frac{1}{2} \mu^{-1/2} \right) + \frac{7\pi^4}{360} T^4 \left(\frac{3}{8} \mu^{-5/2} \right) \right] \\ &= \mu^{3/2} + \frac{\pi^2}{8} T^2 \mu^{-1/2} + \frac{7\pi^4}{640} T^4 \mu^{-5/2}.\end{aligned}\quad (8)$$

Factoring out $\mu^{3/2}$, we obtain the expansion in terms of T/μ :

$$\epsilon_F^{3/2} = \mu^{3/2} \left[1 + \frac{\pi^2}{8} \left(\frac{T}{\mu} \right)^2 + \frac{7\pi^4}{640} \left(\frac{T}{\mu} \right)^4 \right]. \quad (9)$$

(b) To invert the series, we first express μ/ϵ_F as:

$$\frac{\mu}{\epsilon_F} = \left[1 + \frac{\pi^2}{8} \left(\frac{T}{\mu} \right)^2 + \frac{7\pi^4}{640} \left(\frac{T}{\mu} \right)^4 \right]^{-2/3}. \quad (10)$$

Using the binomial expansion $(1+x)^{-2/3} = 1 - \frac{2}{3}x + \frac{5}{9}x^2 + \dots$, we get:

$$\begin{aligned}\frac{\mu}{\epsilon_F} &= 1 - \frac{2}{3} \left[\frac{\pi^2}{8} \left(\frac{T}{\mu} \right)^2 + \frac{7\pi^4}{640} \left(\frac{T}{\mu} \right)^4 \right] + \frac{5}{9} \left[\frac{\pi^2}{8} \left(\frac{T}{\mu} \right)^2 \right]^2 \\ &= 1 - \frac{\pi^2}{12} \left(\frac{T}{\mu} \right)^2 + \left(\frac{5\pi^4}{576} - \frac{7\pi^4}{960} \right) \left(\frac{T}{\mu} \right)^4 \\ &= 1 - \frac{\pi^2}{12} \left(\frac{T}{\mu} \right)^2 + \frac{\pi^4}{720} \left(\frac{T}{\mu} \right)^4.\end{aligned}\quad (11)$$

We need the expansion in terms of $\delta = T/\epsilon_F$. Note that $(T/\mu) = \delta(\epsilon_F/\mu)$. To leading order, $\epsilon_F/\mu \approx 1 + \frac{\pi^2}{12}\delta^2$, which implies:

$$\left(\frac{T}{\mu} \right)^2 \approx \delta^2 \left(1 + \frac{\pi^2}{12} \delta^2 \right)^2 \approx \delta^2 + \frac{\pi^2}{6} \delta^4, \quad \text{and} \quad \left(\frac{T}{\mu} \right)^4 \approx \delta^4. \quad (12)$$

Substituting these back into the expansion gives:

$$\begin{aligned}\frac{\mu}{\epsilon_F} &= 1 - \frac{\pi^2}{12} \left(\delta^2 + \frac{\pi^2}{6} \delta^4 \right) + \frac{\pi^4}{720} \delta^4 \\ &= 1 - \frac{\pi^2}{12} \left(\frac{T}{\epsilon_F} \right)^2 - \frac{\pi^4}{80} \left(\frac{T}{\epsilon_F} \right)^4,\end{aligned}\quad (13)$$

which is the low- T expansion of the chemical potential.

(c) Substitute $D(\epsilon)$ into the internal energy equation:

$$U = \frac{3N}{2\epsilon_F^{3/2}} \int_0^\infty \frac{\epsilon^{3/2} d\epsilon}{e^{\beta(\epsilon-\mu)} + 1}. \quad (14)$$

Apply the Sommerfeld expansion with $\phi(\epsilon) = \epsilon^{3/2}$. Thus, $\Phi(\epsilon) = \frac{2}{5}\epsilon^{5/2}$, $\Phi''(\epsilon) = \frac{3}{2}\epsilon^{1/2}$, and $\Phi^{(4)}(\epsilon) = -\frac{3}{8}\epsilon^{-3/2}$.

$$\begin{aligned}\int_0^\infty \frac{\epsilon^{3/2} d\epsilon}{e^{\beta(\epsilon-\mu)} + 1} &= \frac{2}{5} \mu^{5/2} + \frac{\pi^2}{6} T^2 \left(\frac{3}{2} \mu^{1/2} \right) + \frac{7\pi^4}{360} T^4 \left(-\frac{3}{8} \mu^{-3/2} \right) \\ &= \frac{2}{5} \mu^{5/2} + \frac{\pi^2}{4} T^2 \mu^{1/2} - \frac{7\pi^4}{960} T^4 \mu^{-3/2}.\end{aligned}\quad (15)$$

Multiply by $\frac{3N}{2\epsilon_F^{3/2}}$ and factor out $\mu^{5/2}$:

$$U = N\epsilon_F \left(\frac{\mu}{\epsilon_F} \right)^{5/2} \left[\frac{3}{5} + \frac{3\pi^2}{8} \left(\frac{T}{\mu} \right)^2 - \frac{7\pi^4}{640} \left(\frac{T}{\mu} \right)^4 \right]. \quad (16)$$

- (d) We substitute the result of part (b) into the expression for U . Let $\delta = T/\epsilon_F$. First, evaluate $(\mu/\epsilon_F)^{5/2}$:

$$\begin{aligned} \left(\frac{\mu}{\epsilon_F} \right)^{5/2} &\approx \left(1 - \frac{\pi^2}{12} \delta^2 - \frac{\pi^4}{80} \delta^4 \right)^{5/2} \approx 1 + \frac{5}{2} \left(-\frac{\pi^2}{12} \delta^2 - \frac{\pi^4}{80} \delta^4 \right) + \frac{15}{8} \left(-\frac{\pi^2}{12} \delta^2 \right)^2 \\ &= 1 - \frac{5\pi^2}{24} \delta^2 + \left(-\frac{\pi^4}{32} + \frac{15\pi^4}{1152} \right) \delta^4 = 1 - \frac{5\pi^2}{24} \delta^2 - \frac{7\pi^4}{384} \delta^4. \end{aligned} \quad (17)$$

Next, substitute $(T/\mu)^2 = \delta^2 + \frac{\pi^2}{6} \delta^4$ and $(T/\mu)^4 = \delta^4$ into the bracket:

$$[\dots] = \frac{3}{5} + \frac{3\pi^2}{8} \left(\delta^2 + \frac{\pi^2}{6} \delta^4 \right) - \frac{7\pi^4}{640} \delta^4 = \frac{3}{5} + \frac{3\pi^2}{8} \delta^2 + \frac{33\pi^4}{640} \delta^4. \quad (18)$$

Now multiply the two series up to order δ^4 :

$$\begin{aligned} \frac{U}{N\epsilon_F} &= \left(1 - \frac{5\pi^2}{24} \delta^2 - \frac{7\pi^4}{384} \delta^4 \right) \left(\frac{3}{5} + \frac{3\pi^2}{8} \delta^2 + \frac{33\pi^4}{640} \delta^4 \right) \\ &= \frac{3}{5} + \delta^2 \left(\frac{3\pi^2}{8} - \frac{\pi^2}{8} \right) + \delta^4 \left(\frac{33\pi^4}{640} - \frac{15\pi^4}{192} - \frac{7\pi^4}{640} \right) \\ &= \frac{3}{5} + \frac{\pi^2}{4} \delta^2 - \frac{3\pi^4}{80} \delta^4. \end{aligned} \quad (19)$$

Factoring out $3/5$, we get the internal energy expansion:

$$U = \frac{3}{5} N\epsilon_F \left[1 + \frac{5\pi^2}{12} \left(\frac{T}{\epsilon_F} \right)^2 - \frac{\pi^4}{16} \left(\frac{T}{\epsilon_F} \right)^4 \right]. \quad (20)$$

- (e) For a 3D nonrelativistic gas, $P = \frac{2}{3} \frac{U}{V}$. Therefore, the pressure expansion is:

$$P(n, T) = \frac{2}{5} n\epsilon_F \left[1 + \frac{5\pi^2}{12} \left(\frac{T}{\epsilon_F} \right)^2 - \frac{\pi^4}{16} \left(\frac{T}{\epsilon_F} \right)^4 \right], \quad (21)$$

where $n = N/V$. Recall that $\epsilon_F \propto n^{2/3}$. Distributing $n\epsilon_F$, we have:

$$P \propto \frac{2}{5} n\epsilon_F + \frac{\pi^2}{6} \frac{nT^2}{\epsilon_F} - \frac{\pi^4}{40} \frac{nT^4}{\epsilon_F^3}. \quad (22)$$

Since $n\epsilon_F \propto n^{5/3}$, $n/\epsilon_F \propto n^{1/3}$, and $n/\epsilon_F^3 \propto n^{-1}$, differentiating with respect to n at constant T gives:

$$\begin{aligned} \left(\frac{\partial P}{\partial n} \right)_T &= \frac{5}{3} \left(\frac{2}{5} \epsilon_F \right) + \frac{1}{3} \left(\frac{\pi^2 T^2}{6 \epsilon_F} \right) - (-1) \left(\frac{\pi^4 T^4}{40 \epsilon_F^3} \right) \\ &= \frac{2}{3} \epsilon_F + \frac{\pi^2 T^2}{18 \epsilon_F} + \frac{\pi^4 T^4}{40 \epsilon_F^3} = \frac{2}{3} \epsilon_F \left[1 + \frac{\pi^2}{12} \left(\frac{T}{\epsilon_F} \right)^2 + \frac{3\pi^4}{80} \left(\frac{T}{\epsilon_F} \right)^4 \right] \end{aligned} \quad (23)$$

The isothermal compressibility $\kappa_T = \frac{1}{n}(\partial n / \partial P)_T = [n(\partial P / \partial n)_T]^{-1}$. Expanding the inverse up to order T^4 :

$$\begin{aligned}
 \kappa_T &= \frac{3}{2n\epsilon_F} \left[1 + \frac{\pi^2}{12} \left(\frac{T}{\epsilon_F} \right)^2 + \frac{3\pi^4}{80} \left(\frac{T}{\epsilon_F} \right)^4 \right]^{-1} \\
 &\approx \frac{3}{2n\epsilon_F} \left[1 - \left(\frac{\pi^2}{12} \delta^2 + \frac{3\pi^4}{80} \delta^4 \right) + \left(\frac{\pi^2}{12} \delta^2 \right)^2 \right] \\
 &= \frac{3}{2n\epsilon_F} \left[1 - \frac{\pi^2}{12} \left(\frac{T}{\epsilon_F} \right)^2 - \frac{11\pi^4}{360} \left(\frac{T}{\epsilon_F} \right)^4 \right]. \tag{24}
 \end{aligned}$$

Number fluctuations in a degenerate Fermi gas

Consider a noninteracting Fermi gas in the grand canonical ensemble. For a single one-particle state α , the occupation number satisfies $n_\alpha = 0, 1$.

- (a) Using the grand canonical distribution, show that the number variance of state α is

$$\langle (\Delta n_\alpha)^2 \rangle = \langle n_\alpha \rangle (1 - \langle n_\alpha \rangle) = f(\epsilon_\alpha) [1 - f(\epsilon_\alpha)], \quad (25)$$

where $f(\epsilon)$ is the Fermi–Dirac distribution

$$f(\epsilon) = \frac{1}{e^{\beta(\epsilon - \mu)} + 1}. \quad (26)$$

- (b) Verify the identity

$$f(\epsilon) [1 - f(\epsilon)] = -T \frac{\partial f(\epsilon)}{\partial \epsilon}, \quad (27)$$

and use it to calculate the variance of the total particle number in the grand canonical ensemble:

$$\langle (\Delta N)^2 \rangle = \sum_{\alpha} f(\epsilon_\alpha) [1 - f(\epsilon_\alpha)]. \quad (28)$$

Replacing the sum by an energy integral, show that at low temperature

$$\langle (\Delta N)^2 \rangle = TD(\epsilon_F) + O(T^3). \quad (29)$$

- (c) For a three-dimensional nonrelativistic Fermi gas, use the density of states to show that

$$\frac{\Delta N}{N} \sim \frac{1}{\sqrt{N}} \left(\frac{T}{\epsilon_F} \right)^{1/2}. \quad (30)$$

- (d) Using the grand canonical distribution, show that

$$\langle (\Delta N)^2 \rangle = T \left(\frac{\partial N}{\partial \mu} \right)_{T,V}. \quad (31)$$

For a homogeneous gas, this implies

$$\kappa_T = \frac{1}{n^2} \left(\frac{\partial n}{\partial \mu} \right)_T = \frac{\langle (\Delta N)^2 \rangle}{NnT}. \quad (32)$$

Use the low-temperature result from part (b) to compute κ_T , and verify that it agrees with the leading low-temperature result obtained from the pressure in Exercise 7.5.2.

Solution

- (a) In the grand canonical ensemble, the available configurations for a single noninteracting state α are $n_\alpha = 0$ or 1 . The expected occupation number is:

$$\langle n_\alpha \rangle = \sum_{n_\alpha=0,1} n_\alpha P(n_\alpha) = 1 \cdot P(1) = \frac{e^{-\beta(\epsilon_\alpha - \mu)}}{1 + e^{-\beta(\epsilon_\alpha - \mu)}} = \frac{1}{e^{\beta(\epsilon_\alpha - \mu)} + 1} = f(\epsilon_\alpha). \quad (33)$$

Since n_α can only be 0 or 1, it holds that $n_\alpha^2 = n_\alpha$. Thus:

$$\langle n_\alpha^2 \rangle = \langle n_\alpha \rangle = f(\epsilon_\alpha). \quad (34)$$

The number variance for the state is then:

$$\begin{aligned} \langle (\Delta n_\alpha)^2 \rangle &= \langle n_\alpha^2 \rangle - \langle n_\alpha \rangle^2 = \langle n_\alpha \rangle - \langle n_\alpha \rangle^2 \\ &= \langle n_\alpha \rangle (1 - \langle n_\alpha \rangle) = f(\epsilon_\alpha) [1 - f(\epsilon_\alpha)]. \end{aligned} \quad (35)$$

- (b) We directly differentiate the Fermi-Dirac distribution:

$$\frac{\partial f(\epsilon)}{\partial \epsilon} = \frac{\partial}{\partial \epsilon} \left(e^{\beta(\epsilon - \mu)} + 1 \right)^{-1} = -\frac{\beta e^{\beta(\epsilon - \mu)}}{(e^{\beta(\epsilon - \mu)} + 1)^2}. \quad (36)$$

Observe that $1 - f(\epsilon) = 1 - \frac{1}{e^{\beta(\epsilon - \mu)} + 1} = \frac{e^{\beta(\epsilon - \mu)}}{e^{\beta(\epsilon - \mu)} + 1}$. Therefore:

$$f(\epsilon) [1 - f(\epsilon)] = \frac{1}{e^{\beta(\epsilon - \mu)} + 1} \frac{e^{\beta(\epsilon - \mu)}}{e^{\beta(\epsilon - \mu)} + 1} = -\frac{1}{\beta} \frac{\partial f(\epsilon)}{\partial \epsilon} = -T \frac{\partial f(\epsilon)}{\partial \epsilon}. \quad (37)$$

Because fermions are noninteracting, states are statistically independent. The variance of the total particle number $N = \sum_\alpha n_\alpha$ is the sum of the individual variances:

$$\langle (\Delta N)^2 \rangle = \sum_\alpha \langle (\Delta n_\alpha)^2 \rangle = \sum_\alpha f(\epsilon_\alpha) [1 - f(\epsilon_\alpha)]. \quad (38)$$

Replacing the sum over states with an integral over the energy density of states $D(\epsilon)$, and applying the identity:

$$\langle (\Delta N)^2 \rangle = \int_0^\infty D(\epsilon) \left(-T \frac{\partial f}{\partial \epsilon} \right) d\epsilon. \quad (39)$$

At low temperatures, $-\partial f / \partial \epsilon$ is a sharply peaked function around $\epsilon = \mu$ that approaches a Dirac delta function $\delta(\epsilon - \mu)$. Utilizing the leading term of the Sommerfeld expansion gives:

$$\langle (\Delta N)^2 \rangle = TD(\mu) + O(T^3). \quad (40)$$

Since $\mu = \epsilon_F + O(T^2)$, $D(\mu) = D(\epsilon_F) + O(T^2)$. Therefore, at low temperature:

$$\langle (\Delta N)^2 \rangle = TD(\epsilon_F) + O(T^3). \quad (41)$$

- (c) For a 3D nonrelativistic Fermi gas, the density of states at the Fermi surface is $D(\epsilon_F) = \frac{3N}{2\epsilon_F}$. Substituting this into the result from part (b):

$$\langle(\Delta N)^2\rangle \approx T \frac{3N}{2\epsilon_F}. \quad (42)$$

Taking the square root gives the root-mean-square fluctuation $\Delta N \approx \sqrt{\frac{3N}{2} \frac{T}{\epsilon_F}}$. The relative fluctuation is thus:

$$\frac{\Delta N}{N} \approx \frac{1}{N} \sqrt{\frac{3N}{2} \frac{T}{\epsilon_F}} = \sqrt{\frac{3}{2}} \frac{1}{\sqrt{N}} \left(\frac{T}{\epsilon_F}\right)^{1/2} \sim \frac{1}{\sqrt{N}} \left(\frac{T}{\epsilon_F}\right)^{1/2}. \quad (43)$$

- (d) In the grand canonical ensemble, the partition function is $Z_{GC} = \text{Tr} e^{-\beta(\hat{H}-\mu\hat{N})}$. The expected total particle number is:

$$\langle N \rangle = \frac{1}{\beta Z_{GC}} \frac{\partial Z_{GC}}{\partial \mu} = \frac{1}{\beta} \frac{\partial \ln Z_{GC}}{\partial \mu}. \quad (44)$$

Taking another derivative with respect to μ yields the variance:

$$\frac{1}{\beta} \frac{\partial \langle N \rangle}{\partial \mu} = \frac{1}{\beta^2} \frac{\partial^2 \ln Z_{GC}}{\partial \mu^2} = \langle N^2 \rangle - \langle N \rangle^2 = \langle (\Delta N)^2 \rangle. \quad (45)$$

Thus, replacing $1/\beta$ with T provides $\langle (\Delta N)^2 \rangle = T \left(\frac{\partial N}{\partial \mu} \right)_{T,V}$.

For a homogeneous gas, $N = nV$, so $(\partial N / \partial \mu)_{T,V} = V(\partial n / \partial \mu)_T$. Using the Gibbs-Duhem relation at constant T , $dP = n d\mu$, we have $(\partial P / \partial \mu)_T = n$. By the chain rule:

$$\left(\frac{\partial n}{\partial \mu} \right)_T = \left(\frac{\partial n}{\partial P} \right)_T \left(\frac{\partial P}{\partial \mu} \right)_T = \left(\frac{\partial n}{\partial P} \right)_T n. \quad (46)$$

Substituting this into the fluctuation relation:

$$\langle (\Delta N)^2 \rangle = TVn \left(\frac{\partial n}{\partial P} \right)_T. \quad (47)$$

Since $\kappa_T = \frac{1}{n}(\partial n / \partial P)_T$, we get $(\partial n / \partial P)_T = n\kappa_T$. Substituting this gives $\langle (\Delta N)^2 \rangle = TVn^2\kappa_T = TNn\kappa_T$. Rearranging solves for κ_T :

$$\kappa_T = \frac{\langle (\Delta N)^2 \rangle}{NnT}. \quad (48)$$

Now, using the low-temperature fluctuation $\langle (\Delta N)^2 \rangle \approx T \frac{3N}{2\epsilon_F}$ from part (b):

$$\kappa_T \approx \frac{T \frac{3N}{2\epsilon_F}}{NnT} = \frac{3}{2n\epsilon_F}. \quad (49)$$

In the preceding exercise (Problem 1e), we obtained $\kappa_T = \frac{3}{2n\epsilon_F}[1 + O(T^2)]$. The leading order term exactly perfectly matches $\frac{3}{2n\epsilon_F}$, verifying the fluctuation-compressibility consistency.